



ESP32 — MicroPython Setup and Adhyay1 Library Guide

This file provides a complete step-by-step guide to set up ESP32 with MicroPython and use the Adhyay1 library — from installing MicroPython firmware, adding libraries, to running example code and troubleshooting.

1. Quick Introduction / Requirements

Hardware:

- ESP32 board
- USB Micro-B or USB-C cable (data supported)
- PC (Windows / macOS / Linux)

Software:

- [MicroPython firmware for ESP32](#)
- Adhyay1 Library (downloadable as .mpy file or zip)
- [Thonny IDE](#) or any Python editor with MicroPython support
- Optional: esptool.py (for flashing)
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2. Install MicroPython on ESP32

1. Install [esptool](#) via pip if not already installed:
2. pip install esptool
3. Connect your ESP32 to the PC via USB.
4. Erase the flash:
5. esptool.py --chip esp32 erase_flash
6. Flash the latest MicroPython UF2/firmware .bin file to the ESP32:
7. esptool.py --chip esp32 --port COM3 write_flash -z 0x1000 esp32-20230927-v1.20.1.bin

Replace COM3 with your ESP32 port and firmware filename accordingly.

8. Reset your ESP32 after flashing.

3. Add Adhyay1 Library

1. Connect ESP32 to Thonny or any MicroPython IDE.
2. Create a folder named lib in the ESP32 file system.
3. Copy Adhyay1.mpy or library folder into the lib directory.
4. Your ESP32 is now ready to use Adhyay1 functions.

4. Writing and Running Your First Script

1. Open **Thonny IDE** and select **MicroPython (ESP32)** as the interpreter.
2. Create a new script named main.py.
3. Example using Adhyay1 for LED blink:

```
import time
```

```
from adhyay1 import led
```

```
# Initialize onboard LED (typically GPIO 2)
```

```
led_pin = 2
```

```
while True:
```

```
    led.on(led_pin)
```

```
    time.sleep(1)
```

```
    led.off(led_pin)
```

```
    time.sleep(1)
```

4. Save the script to the root of the ESP32 filesystem. It will run automatically on boot.

5. Using Serial Console

- Open **Thonny** → **View** → **Serial** or **REPL**.
- Use print() statements for debugging.

Example:

```
import time
```

```
while True:
```

```
    print("Hello from Adhyay1")
```

```
    time.sleep(1)
```

6. Installing Additional Libraries (Optional)

- Download MicroPython compatible libraries from [CircuitPython / MicroPython library bundle](#).
- Copy them into the lib folder on ESP32.

7. Common Troubleshooting

- **ESP32 not detected:** Ensure USB cable supports data.
- **Import errors:** Check that Adhyay1.mpy is in lib folder.
- **Script not running:** Ensure script is named main.py and saved on root.
- **Permission issues (Linux):** Try a different USB port or add udev rules.

Your ESP32 is now ready for MicroPython coding with the Adhyay1 library!

